

Two objects, and what they were found to do

Findings on the sound produced by the recovered artefacts B2311.22 and B2311.67, with the questions the survey was unable to close.

Primary	Recovered	Sites	Status
M5.5Ve · 0.0485 AU	2 of 2 catalogued	Kell Rille · Sabik Terminator	Both objects active

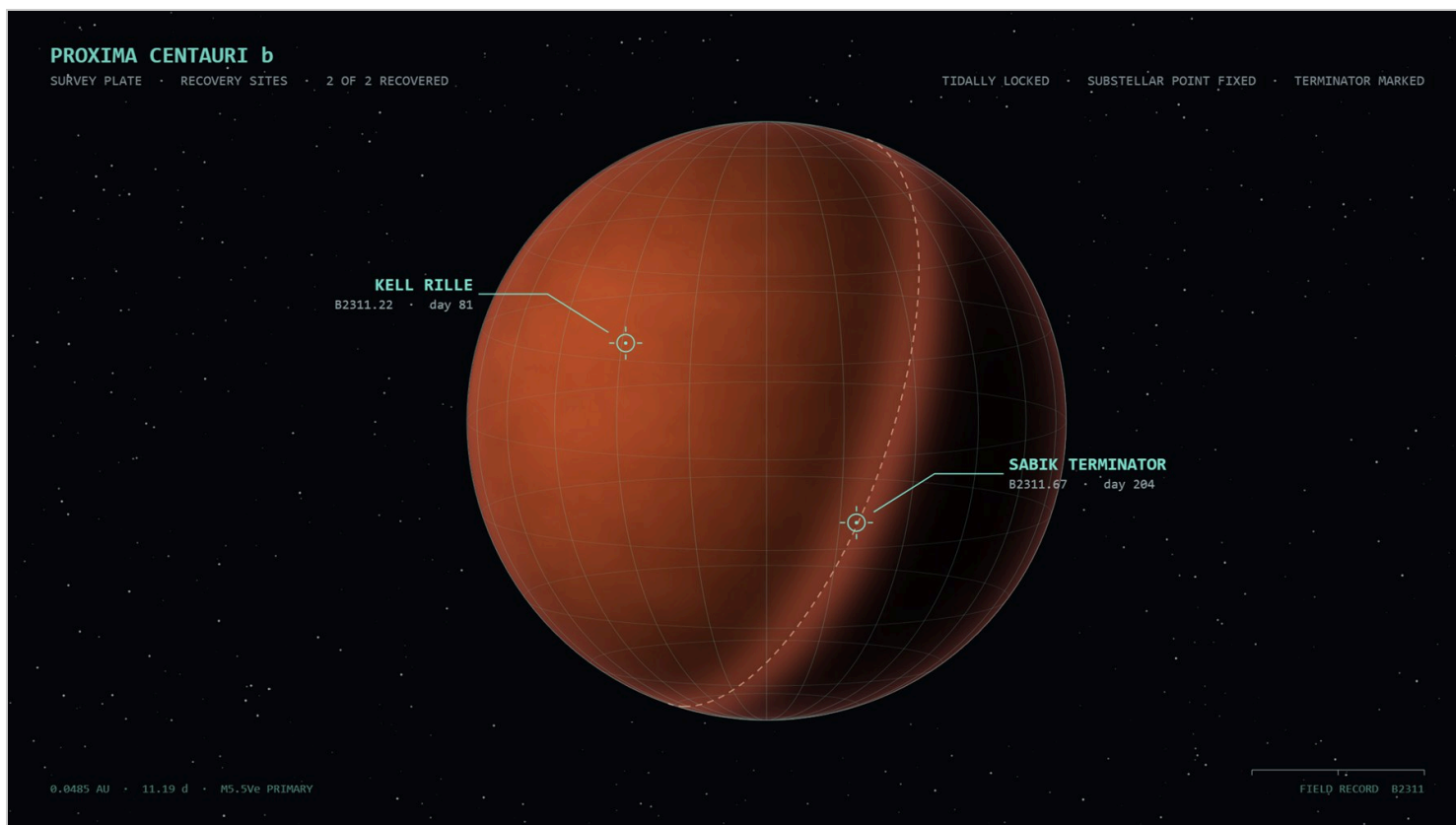
ONE · THE SITUATION

What was found, and in what condition

Neither object was made by us, and neither resembles anything we make. Both were recovered intact, both were sounding when recovered, and neither has been persuaded to stop.

B2311.22 was lifted on day 81 from a dry cistern at Kell Rille, on the lit side. It did not come up alone. A second body lay beside it, and on raising the first, the second changed pitch — an interaction recorded before either had been moved more than a metre, and the first indication that these objects are aware of one another.

B2311.67 was recovered 123 days later from a vault field at the Sabik Terminator, in the permanent twilight where the day side meets the night. There were sixty-seven of them. Each sat at the centre of its own void, spaced with a regularity that survives measurement, and no two voids touched.



Survey plate. The world is tidally locked: the substellar point never moves and the terminator is a permanent ring rather than a passing hour. Kell Rille lies well inside the lit hemisphere; the Sabik vault field lies on the boundary itself.

The first thing established about both objects, and the thing that has shaped every attempt to work with them since, is that *what we are holding is not the whole object*. Each is a body of four spatial dimensions. What the hand meets is a three-dimensional section through it, and what the eye is shown is a two-dimensional projection of that section. Move the section and the object is not altered — a different part of it is simply present. Tissue and structure enter and leave existence at the boundary, continuously, without damage and without any sign that the object regards this as an event.

The second thing established is that *neither object has controls*. There are no markings, no surfaces that invite a hand, nothing that moves when pushed. Everything named in this report — every quantity, every reading, every word on the frames the expedition built around them — is ours. The objects present nothing. The apparatus exists so that a human being can act on them at all, and it should not be mistaken for part of them.

An object whose sound is its own shape

The sound of B2311.22 is the vibration of its own connectedness. It is not struck like a bell or blown like a pipe; it is a network, and what is heard is that network ringing according to how its parts are joined. Change which point is disturbed and the sound changes, because a different part of the structure is being asked to move.

Established. Disturbing the object at its most connected point, versus its most remote, produces spectra differing by 0.51 in relative terms — the same object, two different voices, depending only on where it is touched.

The frequencies it produces do not form any series we use. There is no octave in them and no harmonic ladder; the intervals are those of the object's own construction and nothing else. This was not assumed but measured across the whole catalogue, and it holds without exception: *no specimen sits within twenty cents of any harmonic series*. The closest approach found anywhere in the collection lands exactly on that floor, and no nearer.

Energy does not stay where it is put

The behaviour that most clearly distinguishes this object from anything of ours is what happens after it is disturbed. Our instruments decay: a struck string loses energy and falls silent. B2311.22 does not lose the energy. It *moves* it, along its own structure, from one mode of vibration into its neighbours, so that a sustained note is not a note being held but the object exploring itself.

Established. Over a single sustained disturbance the centre of the sound's energy travels from 645 Hz to 582 Hz without any further input. The object is not decaying. It is redistributing.

Exact recurrence

Under one configuration of the frame the object's output becomes exactly periodic — not approximately, but exactly. Every detail of the sound reassembles itself after an interval between one and a half and four seconds, depending on the specimen, and the reassembly is faithful to a correlation of 0.997. In the interval between recurrences, fractional partial states are audible that belong to neither the beginning nor the end.

The second body

The pair-recovery at Kell Rille was not an accident of storage. When a second object is placed within the same vessel it *responds*, and the character of the response is the most important finding of the survey.

It does not hear everything. It absorbs only at the frequencies it itself possesses, so how well two objects can hear one another is a fixed property of that pair. Two closely related bodies converse fluently; two unrelated ones are very nearly

deaf to each other. When it answers, it answers *in its own frequencies* — the shape of what it heard, re-spoken through a different anatomy. The same statement, in another body's voice.

Established. Ladder overlap between an object and itself measures 1.000; between the least related pair in the catalogue, 0.180. Closely related bodies answer 2.2 times more strongly than unrelated ones — but no pair is ever mute. A darker utterance draws a reply centred at 386 Hz; a brighter one, 406 Hz, from the same responding body.

While both are sounding together, a third set of frequencies appears which belongs to neither of them and cannot be produced by either alone. It exists only for as long as both are sounding, is different for every pair, and vanishes the instant one falls silent. It accounts for roughly a sixth of what is heard during an exchange.

The finding we cannot account for

Being spoken to changes the listener. Not its behaviour — *its structure*. After a sustained exchange the responding body's own frequencies have moved, and they do not move back. The change accumulates in the direction of what it heard.

Established. Thirty seconds of exchange displaced a responding body's own frequencies by 53.9 cents, in nineteen discrete steps, each applied while the object was at rest. The displacement is permanent. An object that has been in a vessel with another for long enough no longer matches its own catalogue entry.

We have no account of the mechanism and no way to reverse it. It should be assumed that every object recovered has already been altered by whatever it spent time beside.

THREE · B2311.67

An object that is ordered and never repeats

The Sabik object is not a body in the sense that B2311.22 is a body. It is a lattice: a regular four-dimensional array, of which we again meet only a section. The section is a tiling of squares and rhombi with eightfold symmetry — perfectly ordered, and never repeating anywhere in its extent.

This is the more difficult of the two objects to describe, because the category its sound belongs to has no name in ordinary use. Sound as we make it is of two kinds. There are tones, in which energy sits at separated frequencies — every string, pipe, bell and oscillator ever built. And there is noise, in which energy is spread continuously — breath, hiss, the sea. Mathematics has always described a third possibility between them, in which energy occupies a set of frequencies that is infinitely subdivided and yet takes up no width at all: a dust with self-similar gaps at every scale.

No material we have ever made produces it. This object does.

Established. Energy injected at one point of the lattice spreads neither as heat spreads nor as a wave travels, but by a power law between the two. The sound arrives smeared in a manner that no room, no enclosure and no material in our possession reproduces.

The lengths at which the structure repeats its own logic are not arbitrary. They follow the sequence 5, 12, 29, 70, 169, 408 — the ratios native to eightfold order, as inevitable to this object as the octave is absent from it.

What moving the section does here

In B2311.22 the section reveals and conceals parts of a body. In B2311.67 it does something else: the lattice *rearranges*. Moving the window causes local reorganisations that propagate outward through the structure, matter appearing and disappearing at the rim, the whole tiling settling into a new arrangement that is equally ordered and equally non-repeating. The object is not disturbed by this. It appears to be the normal condition of such a thing.

The object may also be read in two aspects which appear entirely unrelated — one following the structure directly, one following the pattern its structure would cast — and these have been shown to be the same object described two ways.

FOUR • OPEN

What the survey could not close

Question 1

Why does a listening body change permanently, and is the change legible? The displacement accumulates in the direction of what was heard. If it is a record rather than mere wear, then every recovered object carries the history of whatever stood beside it, and we do not know how to read it.

Question 2

Two objects were found together and interacting. Sixty-seven were found isolated, each at the centre of a deliberate void. Are these two conventions of storage, or two kinds of thing? The vault spacing is too regular to be incidental, and separation is exactly what would prevent the alteration described above.

Question 3

Neither object has an off state. Both were sounding when found, both have sounded continuously since, and the frame can only open onto that sound or close again. What the objects are doing when nothing is listening is not established, and the question may not be well posed.

Question 4

Is B2311.67 a specimen of ordered aperiodic matter, or a device that produces the behaviour of such matter? The distinction matters. The first would make it a sample; the second would make it an instrument, and would mean someone built it to sound like something.

Question 5

We can slide the section along one axis. We cannot rotate it. Everything reported here is what lies in one particular family of cross-sections through each object, and there is no reason to believe it is the informative one.

Question 6

We have found no controls, and have built our own. It has not been established whether the absence of controls is a fact about these objects or a fact about us — whether we are failing to find an interface, or whether the question of an interface does not arise for something that was never meant to be operated.

FIVE • CAUTION

On the frame, and on reading this report

Every quantity named in this report is an expedition quantity. The names on the frames were chosen by us, for our convenience, and describe what our apparatus does to the objects rather than any faculty the objects possess. Where this report says an object "answers" or "explores itself", these are the plainest available descriptions of measured behaviour and are not offered as claims about intent.

What can be said without qualification is that both objects produce sound continuously, that the sound is lawful and repeatable, that it is generated by structures which have no counterpart among our instruments, and that in one case the object is permanently altered by having been listened to and by having listened.

Field record B2311 • Proxima Centauri b • findings compiled from the recovery of B2311.22 (Kell Rille, day 81) and B2311.67 (Sabik Terminator, day 204).

Both objects remain active. Neither has been disassembled.